

EM ALGORITHM AND ITS APPLICATION

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1 Introduction

In real-world machine learning applications, it is common to have many relevant features, but only a subset of them may be observable. When dealing with variables that are sometimes observable and sometimes not, it is indeed possible to utilize the instances when that variable is visible or observed in order to learn and make predictions for the instances where it is not observable. The EM algorithm was proposed and named in a seminal paper published in 1977 by Arthur Dempster, Nan Laird, and Donald Rubin. Their work formalized the algorithm and demonstrated its usefulness in statistical modeling and estimation. The expectation-maximization algorithm can be used to handle situations where variables are partially observable. When certain variables are observable, we can use those instances to learn and estimate their values. Then, we can predict the values of these variables in instances when it is not observable.

1.1 Mathematical Formulation of EM

X is the observed data set.

Y is the not observed data set.

$Z=(X,Y)$ is the complete data set.

$l(\theta;X,Y)=\log$ likelihood of the complete data set.

We wish to obtain MLE(Maximum Likelihood Estimation) of θ .

2 Steps

2.1 E(Expectation) Step

2.1.1 Introduction

In this step, we use the observed data to estimate or guess the values of the missing or incomplete data. It is used to update the variables. Compute the posterior probability or responsibility of each latent variable given the observed data and current parameter estimates. Estimate the missing or incomplete data

values using the current parameter estimates. Compute the log-likelihood of the observed data based on the current parameter estimates and estimated missing data.

2.1.2 Mathematical Formulation of E Step

In this step we compute the expected value of $l(\theta; X, Y)$.

We define, $Q(\theta; \theta_{old}) = E(l(\theta; X, Y) | X, \theta_{old})$

$$= \int l(\theta; X, y) p(y | X, \theta_{old}) dy$$

2.2 M(Maximization) Step

2.2.1 Description

In this step, we need to maximize the likelihood function which we get in the E step. It can be done using several optimization numerical methods.

2.2.2 Mathematical Formulation of M Step

$$\theta = \underset{\theta \in \Theta}{\operatorname{argmax}} Q(\theta; \theta_{old})$$

Now, $l(\theta; x) = \ln(p(X | \theta))$

$$= \ln \int p(X, y | \theta) dy$$

$$= \ln \int [p(X, y | \theta) p(y | X, \theta_{old})] / [p(y | X, \theta_{old})] dy$$

$$= \ln E[[(p(X, Y | \theta)) / (p(Y | X, \theta_{old}))] | X, \theta_{old}]$$

$$\geq E[\ln [(p(X, Y | \theta)) / p(Y | X, \theta_{old})] | X, \theta_{old}]$$

$$= E[\ln(p(X, Y | \theta) | X, \theta_{old}) - E[\ln(p(Y | X, \theta_{old})) | X, \theta_{old}]$$

$$= Q(\theta; \theta_{old}) - E[\ln(p(Y | X, \theta_{old})) | X, \theta_{old}]$$

2.3 Derivation of the EM algorithm

- Suppose we have some data X with joint pdf $p(X | \theta)$
- Let $l(\theta) = \ln p(X | \theta)$ denotes the log - likelihood.
- Now suppose we are trying to guess at θ and improve our guesses through some sort of iteration. Let $\hat{\theta}^{(n)}$ be our n th iteration guess.

- We would like to find a new value for θ that satisfies: -

$$l(\hat{\theta}^{(n)}) \leq l(\theta)$$

- We will introduce some hidden variables Y , either because we are actually working with a model that has hidden(unobserved) variables or "artificially" because they make the maximization more tractable.

- So, we may write,

$$\begin{aligned} l(\theta) - l(\hat{\theta}^{(n)}) &= \ln p(X|\theta) - \ln p(X|\hat{\theta}^{(n)}) \\ &= \ln \left(\int p(X|y, \theta) p(y|\theta) dy \right) - \ln p(X|\hat{\theta}^{(n)}) \\ &= \ln \left(\int \left[\frac{p(X|y, \theta) p(y|\theta)}{p(y|X, \hat{\theta}^{(n)})} \right] p(y|X, \hat{\theta}^{(n)}) dy \right) - \ln p(X|\hat{\theta}^{(n)}) \end{aligned}$$

- Note that the thing in parentheses is an expectation with respect to the distribution of $Y|X, \hat{\theta}^{(n)}$. So, applying Jensen's inequality, we have,

$$\begin{aligned} l(\theta) - l(\hat{\theta}^{(n)}) &\geq \left(\int \ln \left[\frac{p(X|y, \theta) p(y|\theta)}{p(y|X, \hat{\theta}^{(n)})} \right] p(y|X, \hat{\theta}^{(n)}) dy \right) - \ln p(X|\hat{\theta}^{(n)}) \\ &= \left(\int \ln \left[\frac{p(X|y, \theta) p(y|\theta)}{p(y|X, \hat{\theta}^{(n)}) p(X|\hat{\theta}^{(n)})} \right] p(y|X, \hat{\theta}^{(n)}) dy \right) \end{aligned}$$

call the entire integral $d(\theta|\hat{\theta}^{(n)})$.

We then have ,

$$l(\hat{\theta}^{(n)}) + d(\theta|\hat{\theta}^{(n)}) \leq l(\theta)$$

- Note that

$$\begin{aligned} d(\hat{\theta}^{(n)}|\hat{\theta}^{(n)}) &= \int \ln \left(\frac{p(X|y, \hat{\theta}^{(n)}) p(y|\hat{\theta}^{(n)})}{(p(y|X, \hat{\theta}^{(n)}) p(X|\hat{\theta}^{(n)}))} \right) p(y|X, \hat{\theta}^{(n)}) dy \\ &= \int \ln \left(\frac{p(X, y|\hat{\theta}^{(n)})}{p(X, y|\hat{\theta}^{(n)})} \right) p(y|X, \hat{\theta}^{(n)}) dy \\ &= \int \ln(1) p(y|X, \hat{\theta}^{(n)}) dy = 0 \end{aligned}$$

- So, we have that $l(\hat{\theta}^{(n)}) + d(\theta|\hat{\theta}^{(n)})$ is bounded above by $l(\theta)$ and that is equal to this upper bound at the point $\theta = \hat{\theta}^{(n)}$.

If we maximize $l(\hat{\theta}^{(n)}) + d(\theta|\hat{\theta}^{(n)})$ with respect to θ (equivalently maximize $d(\theta|\hat{\theta}^{(n)})$ with respect to θ), we may improve towards maximising $l(\theta)$. We know at least, that we will not get worse.

- Maximizing

$$d(\theta|\hat{\theta}^{(n)}) = \int \ln \left(\frac{p(X|y, \theta) p(y|\theta)}{p(y|X, \hat{\theta}^{(n)}) p(X|\hat{\theta}^{(n)})} \right) p(y|X, \hat{\theta}^{(n)}) dy$$

with respect to θ is equivalent to maximizing

$$\int \ln(p(X|y, \theta) p(y|\theta)) p(y|X, \hat{\theta}^{(n)}) dy$$

with respect to θ .

Note that this is

$$\int \ln(p(X,y|\theta))p(y|X,\hat{\theta}^{(n)})dy$$

which is

$$E_Y[\ln(p(X,Y|\theta))|X,\hat{\theta}^{(n)}]$$

- So, if we could compute this expectation, maximize it with respect to θ , call the result $\hat{\theta}^{(n+1)}$ and iterate, we can improve towards finding the θ that maximizes the likelihood (or at least not get worse). In other words, we can improve towards finding the MLE of θ .

These expectation and maximization steps are precisely the EM algorithm !

3 Importance of EM Algorithm

The Expectation-Maximization (EM) algorithm is defined as the combination of various unsupervised machine learning algorithms, which is used to determine the local maximum likelihood estimates (MLE) or maximum posterior estimates (MAP) for unobservable variables in statistical models. Further, it is a technique to find maximum likelihood estimation when the latent variables are present. It is also referred to as the latent variable model.

A latent variable model consists of both observable and unobservable variables where observable can be predicted while unobserved are inferred from the observed variable. These unobservable variables are known as latent variables. It is also useful to separate two Gaussian mixtures.

4 Application of EM in Brain Tumor Detection

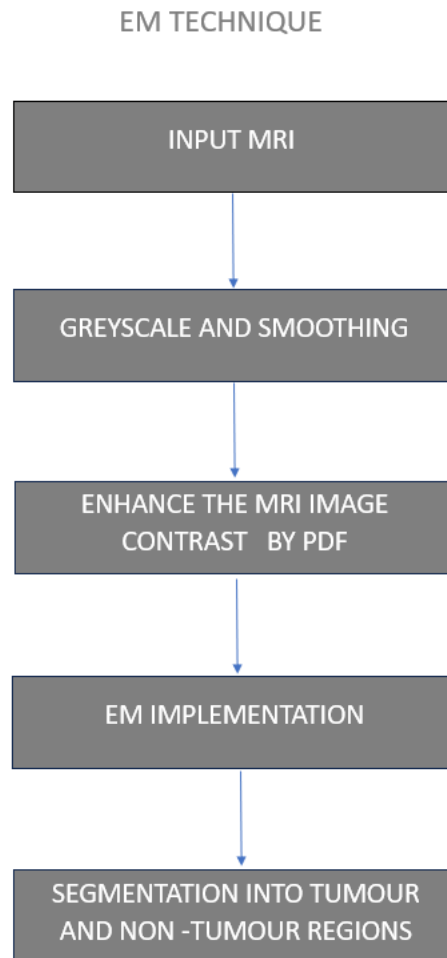
4.1 Introduction

The body is comprised of numerous sorts of cells. Every kind of which has extraordinary capacities. The majority of them develop, afterward partition in a methodical way so new cells are framed. They are expected to retain body function appropriately. These cells and spinal line together shape the Central Nervous System. At the point where the capacity to control their development is lost; separation takes place again and again, with no request. The additional cells shape a lump of the substance called a tumor. Identification of the brain tumor in its beginning time after treatment is the main cure for the sickness. The causes of the ailment are not clear nor are the explanations behind an expanded number of such cases. As of now, there are no techniques to counteract brain tumors, which is the reason early recognition speaks to an essential factor in tumor treatment. MRI is the preferred standpoint over different imaging with delicate tissue yielding a point-by-point image with unambiguous and differentiating limits among anatomical structures. Exact estimations in the analysis are very troublesome because of the various shapes, dimensions and aspects of

tumors. Thus consolidating the strategy named EM (Expectation Maximization) consequently expanding the precision of the framework. So we are going to show the highlighted regions in the tumour in the brain when an MRI image of the brain is given as input.

4.2 Overview of the proposed method

Our method consists of five steps given as follows:



4.2.1 INPUT MRI

To embark on our journey towards tumor detection using MRI images, our foremost task is to curate a comprehensive dataset comprising both tumor and non-tumor samples. This dataset serves as the foundation upon which our algorithm will be trained and evaluated. With meticulous care, we gather MRI images from various sources, ensuring diversity in pathology, imaging parameters, and anatomical regions. Once the dataset is assembled, the crucial step of preprocessing ensues. We meticulously standardize the size, orientation, and resolution of each image, harmonizing the data for seamless integration into our analysis pipeline. Through this diligent process of data collection and preprocessing, we lay the groundwork for our journey towards accurate tumor detection, poised to unlock insights from the intricate world of medical imaging.

4.2.2 GRAYSCALE AND SMOOTHING

Grayscale conversion is a fundamental process in image processing that transforms a color image into a single-channel representation, where each pixel is represented by a single intensity value ranging from black to white. This process simplifies the image, making it easier to analyze and process while preserving important information about the image's structure and features. It reduces each pixel's representation from three color channels (red, green, and blue) to a single intensity value.

The range of 0 to 255 commonly represents the intensity values in an 8-bit grayscale image. In such images, each pixel is represented by a single value ranging from 0 (black) to 255 (white), with intermediate values representing shades of gray. This range encompasses a total of 256 possible intensity levels.

Here's what each value typically represents in an 8-bit grayscale image:

- 0: Represents the darkest shade, usually interpreted as black
- 255: Represents the brightest shade, usually interpreted as white.
- Intermediate values (1 to 254): Represent various shades of gray, with lower values being darker and higher values being lighter.

This range allows for a wide range of intensity levels to be captured and displayed in grayscale images, making them suitable for various image processing and analysis tasks.

Through smoothing operations such as Gaussian filtering or median filtering, we gently sculpt the intensity landscape, blurring high-frequency noise while preserving essential image features. This delicate balance between noise reduction and feature preservation lays the foundation for accurate interpretation and automated analysis of MRI data

4.2.3 ENHANCE THE MRI IMAGE CONTRAST BY PDF

As tumor detection is mainly based on the intensity of MRI images, the contrast of the images is very important. A piecewise linear histogram transformation is usually employed to enhance the intensity contrast.

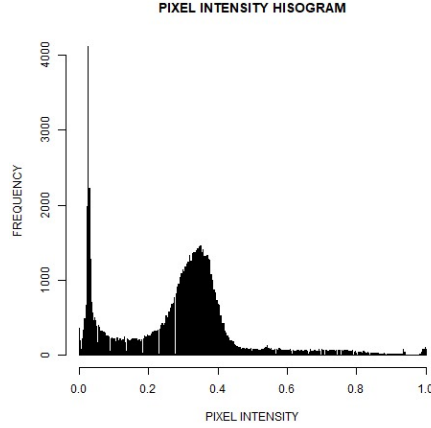


Figure 1: Histogram of intensity vs no of pixel of grayscale MRI image of brain with tumor

However, it is difficult to determine a fixed set of lower and upper limits of the transformation slope for all images because they are data dependent. In this study, we automatically and adaptively determine the parameters from each image. In brain MRI images, there are two classes of tissues: tumor, healthy brain. First, sample voxels of the two classes are manually selected from the training data. The intensity PDFs of each class is estimated.

We also compute their mean values ($\mu_{tumor}^A, \mu_{brain}^A$) their standard deviations ($\sigma_{tumor}, \sigma_{brain}$) and M^A .

where M^A is the intensity value which has the highest probability in brain class.

M^B is the intensity value with highest probability in tumor class. The mean values have the property of ($\mu_{tumor}^A < \mu_{brain}^A$).

As a result, the mean values of the tumor is estimated as

$$\mu_{tumor} = M^B - M^A + \mu_{tumor}^A.$$

Now, we set the lower limit as $T_{min} = \mu_{tumor} - 3\sigma_{tumor}$

4.2.4 EM IMPLEMENTATION

In many applications, we consider a Gaussian Distribution for representing a data distribution because the mathematical manipulations for Gaussian function are relatively easier. In our project, A Gaussian Mixture Model (GMM) is employed to represent the pixel distribution of the main brain tissues. We are interested in finding the probability that each point belongs to a given cluster. The density function can be written as the sum of weighted Gaussians, i.e., suppose we have used 'K' many clusters to represent our data distribution, then density:

$$p(\vec{X}) = \pi_1 N(\vec{X} | \mu_1, \Sigma_1) + \pi_2 N(\vec{X} | \mu_2, \Sigma_2) + \dots + \pi_K N(\vec{X} | \mu_K, \Sigma_K)$$

where $\sum_{j=1}^K \pi_j = 1$ and π_k is called the mixing coefficient for the k th cluster, $0 \leq \pi_k \leq 1$.

This is the mixture model.

Now, the log-likelihood will be:

$$\ln(p(X|\mu, \Sigma, \pi)) = \sum_{n=1}^N \ln(p(X_n)) = \sum_{n=1}^N \ln\{\sum_{j=1}^K \pi_j N(X_n|\mu_j, \Sigma_j)\}$$

where N is the total number of data points.

We cannot apply MLE in this case because there is no closed form solution.

So, we will estimate these parameters using EM algorithm:-

Step 1: Initialize the parameters μ_j, Σ_j, π_j .

Step 2:E-Step

Calculate the posterior probabilities($\gamma_k(X_n)$) based on the current parameter values. Here 'k' is the latent variable.

We can think of mixing coefficients as the prior probabilities for the components.

Posterior probability:

$$\gamma_k(X_n) = p(k|X_n) = \frac{p(k)p(X_n|k)}{p(X_n)} = \frac{\pi_k N(X_n|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(X_n|\mu_j, \Sigma_j)}$$

where $\pi_k = \frac{N_k}{N}$, N_k = Number of points assigned to k th cluster.

Step 3:M-Step

Re-estimate parameter values of the model using the current posterior probabilities. The parameter update formulas are as follows:

$$\begin{aligned} \pi_k^{new} &= \frac{1}{N} \sum_{n=1}^N \gamma_k(X_n) \\ \mu_k^{new} &= \frac{\sum_{n=1}^N \gamma_k(X_n) X_n}{\sum_{n=1}^N \gamma_k(X_n)} \\ \Sigma_k^{new} &= \frac{\sum_{n=1}^N \gamma_k(X_n) (X_n - \mu_k^{new})(X_n - \mu_k^{new})^T}{\sum_{n=1}^N \gamma_k(X_n)} \end{aligned}$$

Repeat the iterations until convergence.

4.2.5 SEGMENTATION INTO TUMOR AND NON-TUMOR REGION

Segmentation, the process of partitioning an image into meaningful clusters, lies at the heart of tumor detection in medical imaging. Grounded in statistical principles, the EM algorithm operates by iteratively estimating the underlying probability distribution of image intensities. Through the interplay of expectation and maximizing steps, the algorithm converges towards a solution that optimally separates tumor and non-tumor components.

The red colour indicates tumor region in (b) of figure 2.

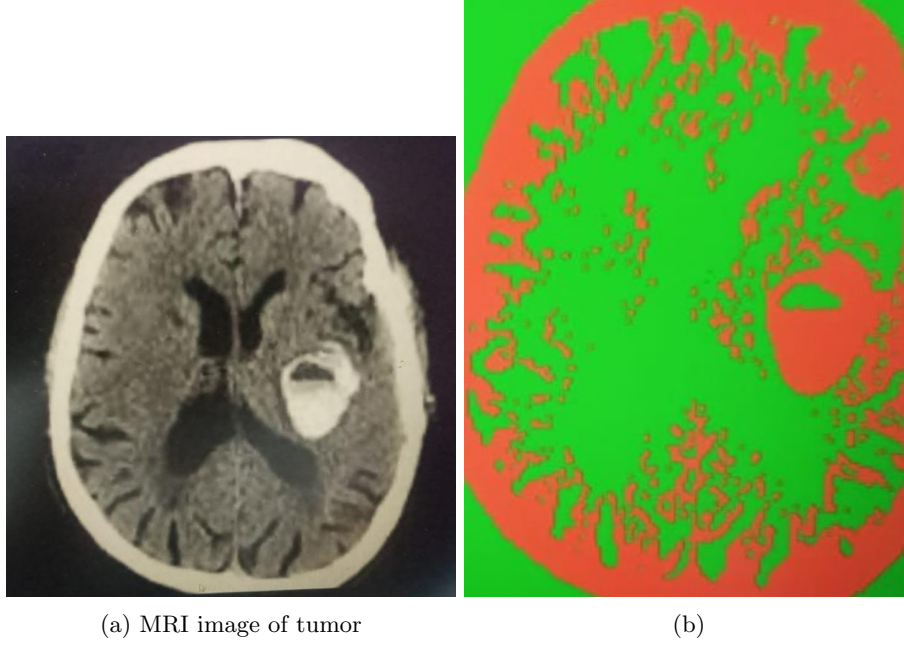


Figure 2: Segmentation into tumor and non-tumor region

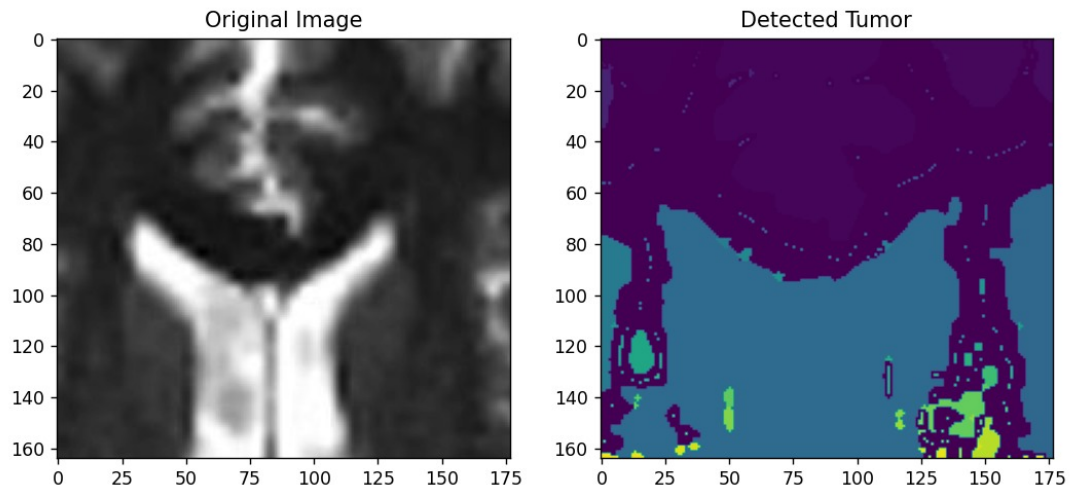


Figure 3: Final Output

Higher colour intensity indicates the tumor regions.

5 LIMITATIONS

While the Expectation-Maximization (EM) algorithm offers notable advantages in segmenting brain tumors from MRI images, it also presents certain limitations:

1. We assume that the underlying probability distributions of image intensities follow a Gaussian distribution (Observing the histogram of intensity and number of pixels). However, this assumption may not always hold true for MRI images, especially in the presence of complex tissue structures, artifacts, or heterogeneous tumor characteristics.

2. MRI images may contain noise and artifacts that can adversely affect the performance of the EM algorithm. Due to the absence of modern software, we are unable to effectively reduce noise.

3. These algorithm require a significant amount of time to differentiate between tumor and non-tumor region.

4. The performance of the EM algorithm may vary across different datasets and tumor types.